PROJECT REPORT

ADMITFLOW

?Admission Conversion Forecasting with Funnel Attribution and Scenario Simulation?

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CERTIFICATE

This is to certify that Samiksha and Zahir, students of T.Y. B.Sc. Data Science, have completed the project titled ?Admission Conversion Forecasting with Funnel Attribution and Scenario Simulation? as part of the BDS-03 Capstone Project for Academic Year 2026?27, subject to verification and approval by the college and project guide.

ABSTRACT

AdmitFlow is a local admission analytics and decision-support prototype developed for the BDS-03 Data Science capstone. It uses a reproducibly generated synthetic dataset of 4,200 applicant records across six admission cycles, 2021?2026. The project calculates a cumulative admission funnel, compares outcomes across cycles and applicant groups, reports descriptive campaign-source associations, evaluates two classifiers for the synthetic enrolled outcome, extrapolates annual enrolments using linear regression, and calculates deterministic what-if scenarios. A React interface presents these outputs through a FastAPI backend. An optional server-side Groq integration answers natural-language questions using aggregate context computed by the API.

The funnel contains 4,200 leads, 3,106 application starts, 1,983 submissions, 1,015 reviews, 708 admissions/offers, and 351 enrolments. Logistic regression and random forest are compared on a stratified random 75/25 split; reported ROC-AUC values are 0.5573 and 0.5195 respectively. The random forest predicts no positive cases at the default threshold, so its 91.62% accuracy does not demonstrate useful detection. Forecasting uses six annual observations, fits on the first four, and evaluates on the last two; its H1/H2/H4/H6 values are 74.2, 77.3, 83.5, and 89.7 enrolments. Scenario outputs are calculated from user-provided multipliers and are not causal predictions. Campaign attribution is descriptive and does not establish campaign impact.

The project demonstrates a traceable analytics workflow, but its synthetic data, weak classifier results, short forecast history, deterministic scenario assumptions, demo-only identity flow, and process-local audit log prevent operational admissions use.

CHAPTER 1 ? INTRODUCTION

Admissions teams need a way to understand where applicants are lost in the funnel, which segments underperform, and how explicit planning assumptions affect projected counts. AdmitFlow brings selected aggregate calculations, model evaluation, a simple forecasting workflow, and deterministic scenario arithmetic into a local analytics platform. The project intentionally distinguishes descriptive observations from causal claims, so the output supports decision support without overstating certainty.

CHAPTER 2 ? SYSTEM ANALYSIS

The project addresses a common academic problem: admissions teams need to see funnel losses and cohort patterns, but many raw datasets do not provide them. The system converts the dataset into a reproducible analytics workflow with validation, trend analysis, decision-relevant reporting, and deterministic planning assumptions.

CHAPTER 3 ? SYSTEM DESIGN

The design follows a layered architecture: data generation and validation, analytics, ML and forecasting, backend API, frontend dashboard, and evidence capture. Each layer remains explicit and independent so the project remains transparent and reproducible.